

Reproducing Temporal Straightening on Medium and Wall

Codex experiment report

July 7, 2026

Abstract

We reproduced temporal straightening with gradient-descent planning on the Medium environment, ran DINO baselines, tested speed-constancy ablations, and completed Wall runs for authors-style straightening and DINO patch features. The Medium reproduction shows equal planner success for authors-style DINO patch cosine straightening and DINO patch without straightening, while DINO CLS performs better. A frozen-DINO speed ablation supports the critique that cosine straightening ignores speed variation, but cannot change measured frozen latent geometry. A trainable DINO-channel adapter ablation addresses that caveat: cosine improves directional straightness while worsening latent speed variation, speed-only improves speed regularity without improving straightness, and cosine plus speed gives the best curvature and mean state distance but not the best planner success. A July 7 follow-up tests Erin’s scale-invariant speed-ratio term and the normalized-acceleration decomposition. Ratio-speed alone improves the speed-constancy diagnostic without directly shrinking latent steps, while normalized acceleration improves curvature and reaches 0.16 planner success, the best DINO-channel adapter success in these one-seed Medium runs. Wall authors and DINO patch runs both reach 0.02 planner success.

1 Motivation

Temporal straightening argues that representation quality can be measured by how easy it makes planning. Pixel observations are encoded into latent states, and a gradient-descent planner optimizes actions in that latent space. The paper encourages local latent trajectories to be directionally straight by minimizing

$$1 - \cos(z_{t+1} - z_t, z_{t+2} - z_{t+1}).$$

This aligns consecutive velocity vectors but removes velocity magnitude. Therefore a latent path can be directionally straight while alternating between fast and slow steps. The paper’s theory assumes constant latent speed, but the cosine loss does not enforce it.

2 Methodology

Medium experiments used 50 generated random-policy episodes of length 100, 20 training epochs, and gradient-descent planning with 50 evaluations and goal horizon 25. The baseline compared authors-style DINO patch cosine straightening, DINO patch without straightening, and DINO CLS without straightening.

The speed ablation added penalties on variation in consecutive latent step magnitudes. The first ablation used frozen DINO patch and CLS features. Because frozen encoders cannot change the measured latent geometry, we then ran a DINO-channel adapter ablation in which the DINO backbone remains frozen but the measured projection and aggregation path is trainable.

The July 7 follow-up replaces direct magnitude smoothness with Erin’s scale-invariant ratio penalty,

$$r_t = \frac{\|v_{t+1}\|}{\|v_t\|}, \quad L_{\text{ratio}} = \left(\sqrt{r_t} - \frac{1}{\sqrt{r_t}} \right)^2.$$

It also tests the normalized-acceleration identity

$$\frac{\|a_t\|^2}{\|v_t\| \|v_{t+1}\|} = L_{\text{ratio}} + 2(1 - \cos \theta_t).$$

Both runs use the trainable `dino_channel` adapter path. Latent diagnostics use a 20-rollout cap because the initial 50-rollout ratio diagnostic behaved like the earlier stalled adapter speed diagnostic.

Wall used the same episode count, episode length, epoch count, and planner settings. The Wall run required code compatibility fixes for missing legacy imports and generated image shape normalization.

3 Medium Results

Variant	Encoder	Loss	Epoch	Train	Val	Success
Authors approach	DINO patch	<code>cos1e-1</code>	20	0.2552	0.2385	0.14
DINO patch	DINO patch	<code>False</code>	20	0.1195	0.1025	0.14
DINO CLS	DINO CLS	<code>False</code>	20	0.0785	0.0691	0.22
Patch speed	DINO patch	<code>aggspeed1e-1</code>	20	0.1268	0.1094	0.14
Patch cosine+speed	DINO patch	<code>cos1e-1+aggspeed1e-1</code>	20	0.2625	0.2455	0.14
CLS speed	DINO CLS	<code>speed1e-1</code>	20	0.0860	0.0754	0.22

Table 1: Completed Medium baseline and frozen-DINO ablation runs.

The frozen patch family had identical measured latent diagnostics across cosine, no-straightening, speed-only, and cosine-plus-speed variants: mean cosine curvature 1.3402, path-to-endpoint ratio 16.0212, latent speed CV 0.3973, and relative speed jump 0.3791. This confirms that frozen-feature ablations are not sufficient to test whether speed constancy can repair the latent geometry.

4 Adapter Ablation

Variant	Success	State dist	Curvature	Speed CV	Speed jump
No straightening	0.10	3.5815	1.3060	0.4577	0.4501
Cosine	0.12	3.3451	1.0271	0.5772	0.6053
Speed-only	0.14	3.4117	1.3431	0.4038	0.3865
Cosine+speed	0.10	3.1176	0.9806	0.4914	0.4911

Table 2: DINO-channel adapter ablation. Speed-only latent diagnostics used 20 rollouts and an epoch-18 checkpoint; other diagnostics used 50 rollouts and epoch 20.

The adapter ablation directly exhibits the cosine-loss caveat. Cosine straightening reduces curvature and path ratio, but increases speed variation. Speed-only reduces speed variation, but

worsens curvature and path ratio. Cosine plus speed gives the best curvature, best path ratio, and best mean state distance, yet it does not improve planner success.

5 Erin Ratio-Speed Ablation

Variant	Train	Val	Success	State dist	Curvature	Ratio penalty
Ratio-speed	0.1426	0.1408	0.12	4.2253	1.3435	0.3153
Normalized acceleration	0.3688	0.3339	0.16	3.5665	1.0340	0.5058

Table 3: July 7 DINO-channel ratio ablations. Both use epoch 20 planning with 50 evaluations. Latent metrics use 20 rollouts.

Variant	Path/end	Norm. accel	Speed CV	Speed jump	p95/p05
Ratio-speed	17.5652	3.0024	0.3793	0.3756	7.7778
Normalized acceleration	13.8773	2.5737	0.4510	0.4598	8.7023

Table 4: Additional latent diagnostics for the ratio ablations. Lower is better for every metric in this table.

Ratio-speed is not the same as the paper author’s smoothness ablation. It does not penalize latent step magnitude directly and it kept much larger latent steps than the earlier magnitude speed-only adapter run. However, ratio-speed alone did not improve directional straightness: curvature 1.3435 is close to the old speed-only adapter value 1.3431 and worse than no-straightening’s 1.3060. Normalized acceleration combines speed ratio and curvature, lowering curvature to 1.0340 and full normalized acceleration to 2.5737. It also reached 0.16 planner success, the best adapter-family success rate observed here. It did not beat the DINO CLS baseline at 0.22, and its pure speed metrics were worse than ratio-speed alone.

6 Wall Results

Variant	Encoder	Loss	Epoch	Train	Val	Success
Authors approach	DINO patch	<code>cos1e-1</code>	20	0.2824	0.2715	0.02
DINO patch	DINO patch	<code>False</code>	20	0.1385	0.1209	0.02

Table 5: Completed Wall runs with 50 planner evaluations. Both variants had mean state distance 33.4643.

Wall DINO without straightening trains to a lower validation loss than the authors-style cosine run, but both produce the same planner outcome: success rate 0.02, mean state distance 33.4643, mean visual distance 1.1894, and mean proprio distance 21.4001.

7 Image Evidence

Patch speed and patch cosine-plus-speed Medium contact sheets are byte-identical, while CLS contact sheets differ from patch contact sheets. This aligns with the frozen-latent finding: unchanged

frozen measured geometry yields unchanged image evidence, while changing encoder family changes the images and planner metrics. The Wall planner path produced logs and plan targets but no rendered image/contact-sheet media.

8 Discussion

The experiments support the critique that cosine temporal straightening alone does not enforce the constant-speed condition used in the theoretical story. The adapter ablation separates directional straightness from speed regularity and shows that improving one does not automatically improve the other. Erin’s ratio term addresses the author’s smoothness concern more directly because it is scale-invariant and does not directly shrink velocities. The normalized-acceleration variant is the most promising new result, but planner success does not monotonically follow the straightness or speed metrics in these small runs. More seeds and larger datasets would be needed before claiming the speed-ratio penalty is a robust improvement.

9 Artifacts

Primary evidence files are under `modal_evidence/`. Key files include:

- `medium-full-20260702-01/combined_results.json`
- `medium-full-20260702-01/latent_analysis.json`
- `medium-ablations-20260702-01/combined_medium_results.json`
- `medium-ablations-20260702-01/image_evidence_summary.json`
- `medium-adapter-ablations-20260702-01/adapter_results.json`
- `medium-ratio-ablations-20260707-01/ratio_results.json`
- `medium-ratio-ablations-20260707-01/README.md`
- `wall-full-20260703-01/wall_results.json`

Large checkpoint archives and planner target pickles remain local or on the Modal volume and are not included in the PR.